Unit: 6 Unit 8: Quadratics **Name:** ______________
Topic: Graphing Quadratics and Properties of Parabolas **Date:** __________

Advanced (Habanero)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?
(A) $(-2, 0)$
(B) $(-2, 4)$
(C) $(0, 4)$
(D) $(2, 0)$
(E) $(0, 0)$

Q2. Which of the following is the axis of symmetry for the parabola $y = x^2 - 6x + 8$?
(A) $x = -2$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$

Q3. What are the x-intercepts of the parabola $y = x^2 - 4x - 3$?
(A) $(-1, 0), (3, 0)$
(B) $(-3, 0), (1, 0)$
(C) $(-2, 0), (2, 0)$
(D) $(-4, 0), (0, 0)$
(E) $(-1, 0), (5, 0)$

Q4. What is the y-intercept of the parabola $y = 2x^2 - 5x - 3$?
(A) $(0, -3)$
(B) $(0, 3)$
(C) $(0, 2)$
(D) $(0, -2)$
(E) $(0, 1)$

Q5. Which parabola opens downward and has a vertex at $(1, 2)$?
(A) $y = -(x - 1)^2 + 2$
(B) $y = (x - 1)^2 + 2$
(C) $y = -(x + 1)^2 - 2$
(D) $y = (x + 1)^2 - 2$
(E) $y = x^2 - 2$

Q6. What is the minimum value of the function $y = x^2 + 2x + 1$?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q7. What is the maximum value of the function $y = -x^2 + 4x - 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q8. Which equation represents a parabola with an axis of symmetry at $x = -2$?
(A) $y = x^2 - 4x + 3$
(B) $y = x^2 + 4x + 3$
(C) $y = x^2 + 2x + 1$
(D) $y = x^2 - 2x - 3$
(E) $y = x^2 + 4x - 3$

Open Ended Questions (FRQ)

Q9. Graph the parabola $y = x^2 - 3x - 4$ and identify the vertex, axis of symmetry, and x-intercepts.

Q10. Find the equation of the parabola that has a vertex at $(2, 1)$, an axis of symmetry at $x = 2$, and passes through the point $(0, -3)$.

Chef's Tips

- Ensure that students use the correct formula for finding the vertex and axis of symmetry.
- Encourage students to check their work by plugging their answers back into the equation.
- Remind students that graphing parabolas requires attention to detail and an understanding of the properties of parabolas.